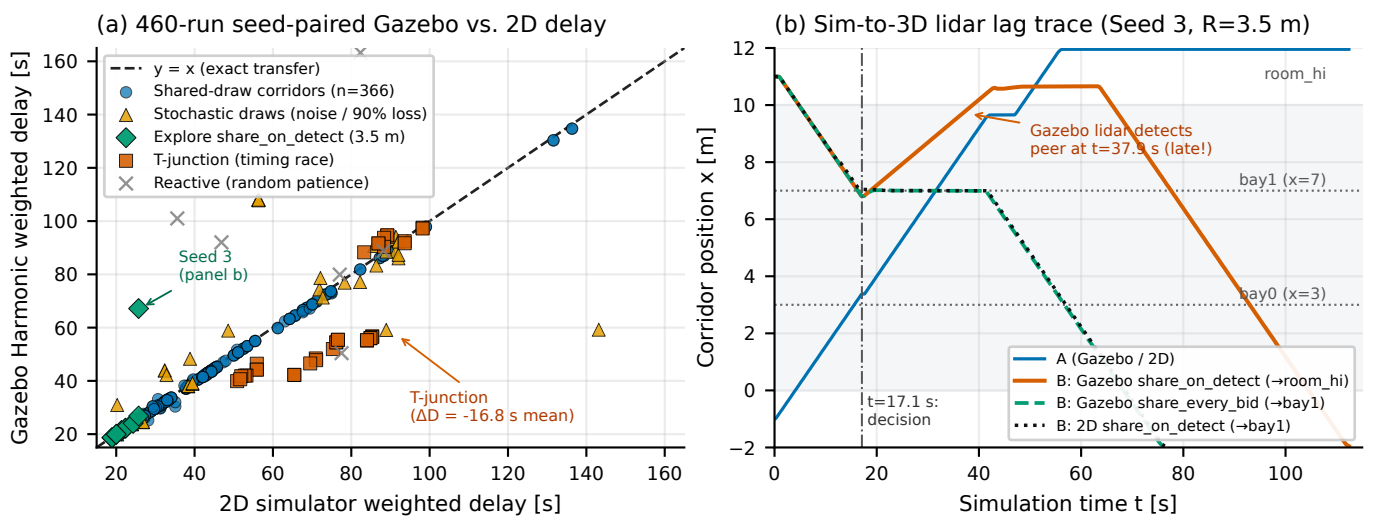
(a) 460-run seed-paired Gazebo vs. 2D delay
y = x (exact transfer)
Shared-draw corridors (n=366)
Stochastic draws (noise / 90% loss)
Explore share_on_detect (3.5 m)
T-junction (timing race)
Reactive (random patience)
Seed 3
(panel b)
T-junction
(ΔD = -16.8 s mean)
Gazebo Harmonic weighted delay [s]
2D simulator weighted delay [s]
(b) Sim-to-3D lidar lag trace (Seed 3, R=3.5 m)
room_hi
Gazebo lidar detects
peer at t=37.9 s (late!)
bay1 (x=7)
bay0 (x=3)
A (Gazebo / 2D)
B: Gazebo share_on_detect (→room_hi)
B: Gazebo share_every_bid (→bay1)
B: 2D share_on_detect (→bay1)
t=17.1 s:
decision
Corridor position x [m]
Simulation time t [s]